

PostgreSQL Setup with Docker

The screenshot shows the Docker Desktop application window. The top bar includes the Docker logo, a search bar, and a 'Sign' button. The left sidebar contains navigation options: Gordon, Containers (selected), Images, Logs, Volumes, Kubernetes, Builds, Models, MCP Toolkit (marked BETA), Docker Hub, and Extensions. The main area displays the 'postgres1' container, which is running. The 'Logs' tab is active, showing a detailed log of the PostgreSQL initialization and startup process. The logs indicate that the database system was successfully started and is now ready for connections. The bottom status bar shows system metrics: RAM 0.73 GB, CPU 0.20%, and Disk 4.61 GB used (limit 58.37 GB). A 'Terminal' button and an 'Update available' notification are also visible.

Containers / postgres1

postgres1
d1e379c76992 postgres:16
5432.5432

STATUS
Running (2 seconds ago)

Logs Inspect Bind mounts Exec Files Stats

Success. You can now start the database server using:

```
pg_ctl -D /var/lib/postgresql/data -l logfile start
```

initdb: warning: enabling "trust" authentication for local connections
initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.
waiting for server to start....2026-09-01 21:40:09.842 UTC [48] LOG: starting PostgreSQL 16.15 (Debian 16.15-1.pgdg13+2) on aarch64-unknown-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
2026-09-01 21:40:09.843 UTC [48] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-09-01 21:40:09.847 UTC [51] LOG: database system was shut down at 2026-09-01 21:40:09 UTC
2026-09-01 21:40:09.849 UTC [48] LOG: database system is ready to accept connections
done
server started

/usr/local/bin/docker-entrypoint.sh: ignoring /docker-entrypoint-initdb.d/*

waiting for server to shut down....2026-09-01 21:40:09.959 UTC [48] LOG: received fast shutdown request
2026-09-01 21:40:09.960 UTC [48] LOG: aborting any active transactions
2026-09-01 21:40:09.961 UTC [48] LOG: background worker "logical replication launcher" (PID 54) exited with exit code 1
2026-09-01 21:40:09.961 UTC [49] LOG: shutting down
2026-09-01 21:40:09.962 UTC [49] LOG: checkpoint starting: shutdown immediate
2026-09-01 21:40:09.964 UTC [49] LOG: checkpoint complete: wrote 3 buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.001 s, sync=0.001 s, total=0.004 s; sync files=2, longest=0.001 s, average=0.001 s; distance=0 kB, estimate=0 kB; lsn=0/14F2AD0, redo lsn=0/14F2AD0
2026-09-01 21:40:09.966 UTC [48] LOG: database system is shut down
done
server stopped

PostgreSQL init process complete; ready for start up.

2026-09-01 21:40:10.078 UTC [1] LOG: starting PostgreSQL 16.15 (Debian 16.15-1.pgdg13+2) on aarch64-unknown-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
2026-09-01 21:40:10.079 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
2026-09-01 21:40:10.079 UTC [1] LOG: listening on IPv6 address ":::", port 5432
2026-09-01 21:40:10.080 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-09-01 21:40:10.082 UTC [62] LOG: database system was shut down at 2026-09-01 21:40:09 UTC
2026-09-01 21:40:10.084 UTC [1] LOG: database system is ready to accept connections
2026-09-01 21:42:57.140 UTC [1] LOG: received fast shutdown request
2026-09-01 21:42:57.142 UTC [1] LOG: aborting any active transactions
2026-09-01 21:42:57.142 UTC [74] FATAL: terminating connection due to administrator command
2026-09-01 21:42:57.143 UTC [1] LOG: background worker "logical replication launcher" (PID 65) exited with exit code 1
2026-09-01 21:42:57.144 UTC [60] LOG: shutting down
2026-09-01 21:42:57.145 UTC [60] LOG: checkpoint starting: shutdown immediate
2026-09-01 21:42:57.152 UTC [60] LOG: checkpoint complete: wrote 44 buffers (0.3%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.003 s, sync=0.002 s, total=0.008 s; sync files=11, longest=0.002 s, average=0.001 s; distance=260 kB, estimate=260 kB; lsn=0/1533E80, redo lsn=0/1533E80
2026-09-01 21:42:57.155 UTC [1] LOG: database system is shut down

PostgreSQL Database directory appears to contain a database; Skipping initialization

2026-09-01 21:43:18.846 UTC [1] LOG: starting PostgreSQL 16.15 (Debian 16.15-1.pgdg13+2) on aarch64-unknown-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
2026-09-01 21:43:18.847 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
2026-09-01 21:43:18.847 UTC [1] LOG: listening on IPv6 address ":::", port 5432

Engine running | | : RAM 0.73 GB CPU 0.20% Disk: 4.61 GB used (limit 58.37 GB)

_ Terminal Update available

Python Web App with Docker Compose

The screenshot shows the Docker Desktop interface. On the left is a sidebar with navigation options: Gordon, Containers (selected), Images, Logs, Volumes, Kubernetes, Builds, Models, MCP Toolkit (BETA), Docker Hub, and Extensions. The main panel displays the 'Containers' view for a container named 'postgres1'. The container is running, as indicated by the 'STATUS' bar which says 'Running (2 seconds ago)'. Below the container name, there are tabs for 'Logs', 'Inspect', 'Bind mounts', 'Exec', 'Files', and 'Stats'. The 'Logs' tab is active, showing a detailed log of the PostgreSQL initialization process. The log includes messages such as 'Success. You can now start the database server using:', 'pg_ctl -D /var/lib/postgresql/data -l logfile start', and various log entries from the PostgreSQL server, including warnings about 'trust' authentication, startup logs, and shutdown logs. At the bottom of the interface, a status bar shows 'Engine running' and system resource usage: 'RAM 0.73 GB CPU 0.20% Disk: 4.61 GB used (limit 58.37 GB)'. There are also buttons for 'Terminal' and 'Update available'.

<https://github.com/eacruz18/ITCS6190-H3-Ed-Cruz/tree/development>

